FMKT 256 EXTRA CREDIT, PROMPT 1

Founder's Five

A CONCENTRATED-STOCK REPLAY

PLAY IT IN A BROWSER. NO LOGIN.

https://pjogilvie7.github.io/founders-five/

Scan the code with a phone, or click the link. Open the link in a private window to see what a first-time visitor sees. Click **Watch a finished run** to reach a scoreboard in about 15 seconds.



The title screen. The strip under the buttons shows the data-as-of date: prices through Mon Nov 2, 2026 close, updated nightly, SpaceX card at 99 trading days.

Founder's Five is a browser game in which you hold $100 million of one real stock on its IPO day and cannot sell it yet. You work through the real lockup release dates, trading windows and SEC rules using only a private bank's tools: staged sales, a 10b5-1 plan, a zero-cost collar, a credit line against the shares and an index fund. Each run is scored on Sharpe ratio, max drawdown, after-tax IRR and after-tax family wealth, against three rival bots that play the same card.

1 + 6 cards: the live SpaceX card and 6 blind five-year IPOs from 2019 to 2021	**99** SpaceX trading days so far; one more is added each trading day	**4** measures: Sharpe, max drawdown, after-tax IRR, after-tax wealth	**185** automated tests pass, including 12 numbered golden tests

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HOW IT MEETS PROMPT 1

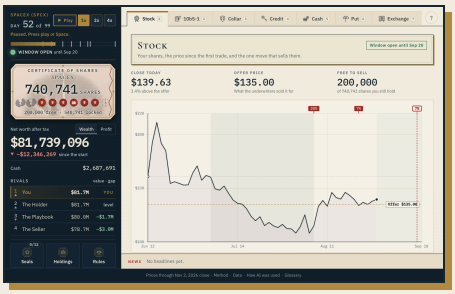
The assignment, point by point

Each requirement from the prompt, what was built to meet it, and where you can see it.

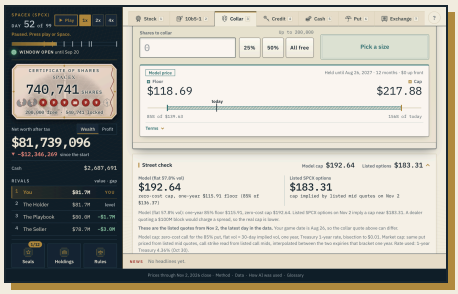
REQUIREMENT	HOW FOUNDER'S FIVE MEETS IT	WHERE YOU SEE IT
Build a finance app using Claude or ChatGPT	Built end to end with Claude Code. Claude also read the prospectus lockup terms and wrote the anonymized headlines, and each of those outputs passed a machine check. The finished game calls no AI model.	"How AI was used" page; PDF page 4
Web browser interface	A static site on GitHub Pages: plain HTML, JavaScript and JSON, nothing to install. Playable on a phone in portrait.	The link on page 1
Project type: trading game	One stock plus its tools (staged sales, 10b5-1 plan, zero-cost collar, protective put, credit line, exchange fund, index fund). A fixed $100M stake over five years, judged on Sharpe ratio, max drawdown and after-tax IRR, the measures the prompt names. Three rival bots play the same card.	Scoreboard; PDF page 5
Imports historical data	Daily prices for the SpaceX card and 6 blind IPO cards, the S&P 500 fund (SPY), three Treasury and SOFR rate series, 8-K earnings dates and each prospectus lockup, all committed as JSON.	Data page; Method page
Updates when new data arrive	A GitHub Action runs on weekdays at 23:30 UTC. It appends prices, rates, SpaceX option data and new SEC filings, runs the tests, and commits only if they pass. The SpaceX card grows by one day each trading day.	Data-as-of strip on every screen; update log on the Data page
No login, no email request	Public site with no accounts and no email prompt. It loads fonts and a chart library from public CDNs and nothing else.	Open the link in a private window on a phone
PDF in Canvas with embedded links	This 5-page PDF. The live site, the Method and Data pages, the Actions history and the repository are clickable.	This file
Accuracy ("no partial credit")	185 tests pass under `node --test`, including 12 numbered golden tests. A spreadsheet with live formulas re-derives 6 of them. Every lockup was read twice by separate agents and reconciled against the prospectus text. Every enforced rule links its source.	Method page; test badge on the Data page; `tests/golden.xlsx`
On time	Due Tuesday, November 3 at 1 pm. The PDF and the live site go in together.	Canvas submission

A grader's first minute

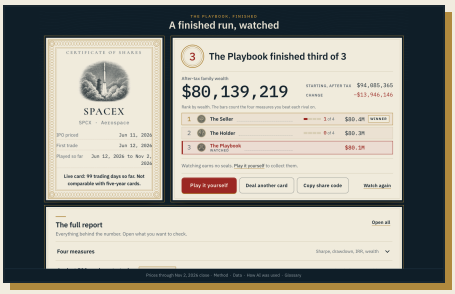
1. Open the link. The title shows the headline, three buttons and the data-as-of strip.
2. Click **Watch a finished run**. The SpaceX card plays The Playbook at 8x, about 15 seconds.
3. The scoreboard opens: four measures for The Playbook against The Holder and The Seller.
4. Click **Play it yourself** or **Deal a blind IPO**. Method and Data are in the footer.



Play screen, SpaceX card, mid-game.



Collar tab with the Street check.

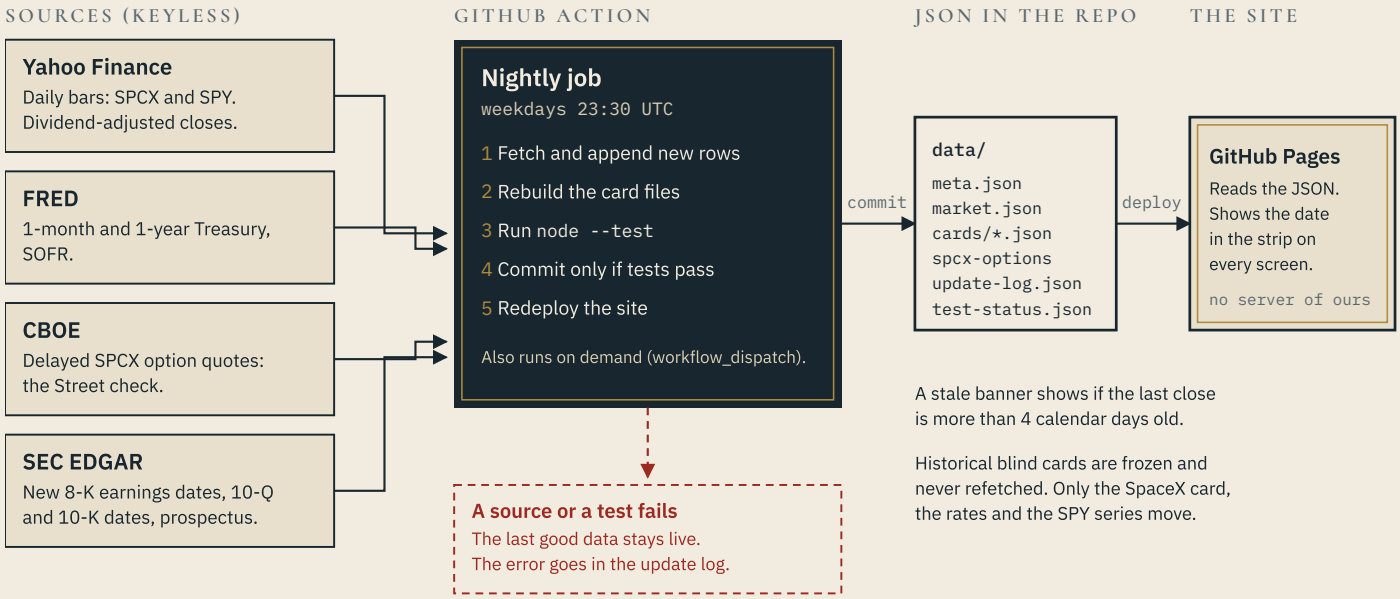


Scoreboard after "Watch a finished run".

HOW THE DATA UPDATES

Nightly, from the sources to the page

The browser never calls a data source. A scheduled job fetches everything, tests it and commits plain JSON. The site only reads that JSON, so a visitor needs no key, no login and no server of ours.



Fetch

Prices, rates, option quotes and filing dates come from free public endpoints with no key. SEC requests carry a contact user agent and stay under ten a second.

Test

The job runs the full test suite (185 tests now) before it commits. If a source fails or a test fails, nothing new goes live and the error is written to the update log.

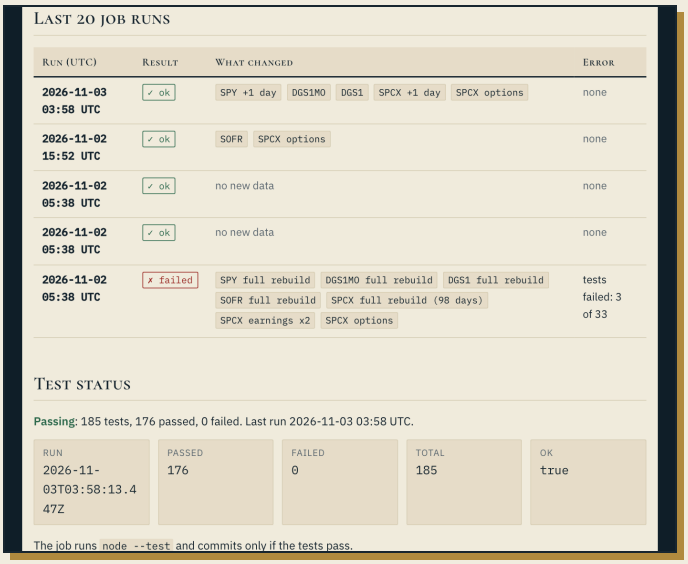
Show

Every screen carries the data-as-of date. The copy says “nightly”, never an exact time, because GitHub’s scheduler can run late.

Check it yourself

- The Data page: data-as-of date, last 20 job runs, test status, links to every JSON file.
- The Actions history on GitHub: every run, its log and the commit it made.
- The repository: the workflow files in `.github/workflows` and the data scripts in `scripts`.

The screenshot shows the update log as it stood when this PDF was built, November 3, 2026: data as of Mon Nov 2, 2026, SpaceX card at 99 trading days.



Last 20 job runs

Run (UTC)	Result	What changed	Error
2026-11-03 03:58 UTC	ok	SPY +1 day · DGS1MO · DGS1 · SPCX +1 day · SPCX options	none
2026-11-02 15:52 UTC	ok	SOFR · SPCX options	none
2026-11-02 05:38 UTC	ok	no new data	none
2026-11-02 05:38 UTC	ok	no new data	none
2026-11-02 05:38 UTC	failed	SPY full rebuild · DGS1MO full rebuild · DGS1 full rebuild · SOFR full rebuild · SPCX full rebuild (98 days) · SPCX earnings x2 · SPCX options	tests failed: 3 of 33

Test status

Passing: 185 tests, 176 passed, 0 failed. Last run 2026-11-03 03:58 UTC.

Run	Passed	Failed	Total	Ok
2026-11-03T03:58:13.447Z	176	0	185	true

The job runs `node --test` and commits only if the tests pass.

The Data page: the update log and the test status.

What Claude did and how we checked it

All AI work ran inside the build sessions. Counts come from the files named under each heading.

9	64	17	185	0
prospectuses read twice by separate agents (18 reads)	anonymized headlines written; 4 identification rounds	findings from the banker and quant critics	tests pass, with a spreadsheet cross-check	AI calls while you play; the game is static

The build

`PROGRESS.md`

Claude Code ran the build from a written spec. One orchestrating session split the work into waves and gave pieces to parallel subagents: the data job, the card universe, the engine, the screens, the pages, the art and the tests. A progress file records each task, its owner, its check and its result, and every overnight decision is logged with its reason and what it costs if wrong.

Lockups, read twice

`data/src/*/lockup*.json`

The SpaceX prospectus and 8 blind-card prospectuses (9 filings) were each read by two agents with no shared context. Each wrote the tranches with a quoted sentence and a page number. A reconciler then compared the two reads and settled every difference against the filing text. Code checks that percentages add to 100 per group and that dates resolve. The two reads differed on 4 items: two date conventions on SpaceX, one release group on Card 4 and one failed price test on Card 6. Each was settled by quoting the filing. In two further cases the reconciler overruled a reading on a more specific source: SpaceX day 180 stays December 8, and one unshipped blind card's lockup end follows the company's own 8-K. All 8 blind cards reconciled and 6 shipped.

Headlines without names

`scripts/check-headlines.mjs`

Each blind card shows 8 headlines taken from its 8 largest one-day moves. Claude wrote them from the 8-K text with no company, product, person, place or ticker. A name filter built from the company name, ticker and the executive and product names in the prospectus rejects any line that contains a banned term. A second agent was then asked to name the company, over 4 identification rounds. Four of the 6 shipped cards ended up unidentified. An expert reader can still name Card 2 (85% confidence) and weakly Card 4 (60%); both are logged, and a swap candidate is ready.

Reveal paragraphs

`data/src/*/reveal.json`

Each blind card ends with a short paragraph about what happened. A script checks that every number in it appears in the card's own data, or is a rounding of a value derived from it.

Accuracy critics

`research/critic-banker, critic-quant`

Two separate agents tried to break the build: a banker critic on the rules (9 findings) and a quant critic on the numbers (8). The quant critic found the core clean: net worth recomputed from its parts, Sharpe, drawdown and IRR recomputed for every bot, card and seat, no look-ahead, identical results over 39 repeat runs. What they caught, now fixed or logged as stated game terms:

- The founder's 10b5-1 cooling-off used the wrong quarterly report; it now follows 17 CFR 240.10b5-1(c)(1)(ii)(B).
- The game allowed a new hedge while a 10b5-1 plan was active, which the rule forbids.
- The margin-call deadline could not be met on its stated date; a forced sale was sized on after-tax proceeds.
- Skipping the $10M house payment changed scores, so the skip option was removed.
- Terminal tax ignored realized losses, and collars could open before 20 days of price history.
- The Method page said returns used adjusted closes while the engine dropped dividends; dividends are now credited and taxed.

Golden tests and the spreadsheet

`engine/*.test.js, tests/golden.xlsx`

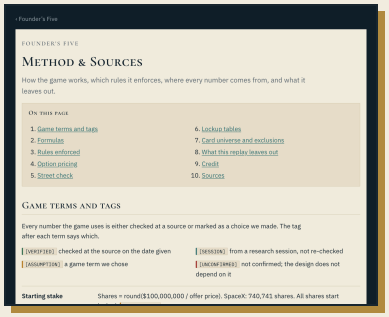
185 tests pass under `node --test`. Twelve are numbered golden tests with known answers: Black-Scholes and the zero-cost call, put-call parity, XIRR, Sharpe, max drawdown, the SpaceX ladder, both cooling-off rules, the loan call, tax, the trading window, determinism and daily invariants. Claude also built `tests/golden.xlsx`, which re-derives six of them with live Excel formulas; its summary sheet compares each cell with the JavaScript value. The nightly job runs the suite and commits only if it passes. The Method page shows each tranche's human sign-off status next to its prospectus quote and page.

METHOD

How the replay is scored, and what binds it

Four measures

Net worth	W = shares x close + collar and put marks + index fund + exchange fund + cash − loan − overdraft − tax owed
Sharpe ratio	mean(R − rf) / stdev(R − rf) x √252, daily return R adjusted for spending, rf from the 1-month Treasury. One decimal.
Max drawdown	Largest peak-to-trough fall of the spending-adjusted wealth index, with both dates.
After-tax IRR	XIRR on day-0 wealth, each spending date and final wealth less the tax still to come (23.8% on gains). Day 0 is also taken after tax, so the tax already inside the stake is not charged to your choices.
After-tax family wealth	The final flow plus total spending. It is the big number; there is no composite grade.



The Method page carries every formula verbatim and a source for every rule.

Rules the engine enforces

Rule	What the game does	Source
Lockup	Shares release only on resolved tranche dates; locked shares cannot be sold, hedged or pledged.	Each card's 424B4, quoted with page; SpaceX 424B4
Trading window	Closed from 30 days before an earnings release until the second trading day after it.	Game term; real 8-K Item 2.02 dates
10b5-1 cooling-off	Engineer 30 days. Founder: the later of 90 days or 2 business days after the quarter's report, at most 120.	17 CFR 240.10b5-1(c)(1)(ii)(B)
No hedge under a plan; one plan	No new collar or put while a plan is adopted; a second plan is blocked.	240.10b5-1(c)(1)(i)(C) and (c)(1)(ii)(D)
Rule 144 (founder)	Volume cap per rolling 3 months; no sales before IPO day 91.	17 CFR 230.144(c), (d), (e)
Single-stock lending	25% advance, 35% call, deadline on the third trading day, then a taxable forced sale.	Game terms; FINRA
Collar band	Put at most 90%, call at least 110% of the price.	Game term; IRC 1259 has no safe harbor
Exchange fund	Seven-year hold.	IRC 704, 737 (secondary sources)

Pricing: Black-Scholes with the 1-year Treasury rate. SpaceX's Street check compares the model's zero-cost cap with listed options. Elsewhere volatility is 60-day realized volatility times 1.10, floored at 30%, with a 1.5 point skew. Each Method-page term is tagged verified, session, assumption or unconfirmed.

What the replay leaves out

- Pre-IPO GRATs, gifts to trusts and charitable remainder trusts
- Donor-advised funds
- QSBS (Section 1202)
- State tax
- Straddle rules; collar settlements are taxed at 23.8% when they settle
- Real dealer spreads; options are struck at the model mid price
- Leaving an exchange fund early
- Any view on what to do with SpaceX or any stock today

A historical replay for education. Not investment, tax or legal advice, and not a view on any stock today.